

CLASS XII · LECTURE 15

Paper II Drill

Both options · One sitting each · Do only the option you will enter in the examination

By the end of this lecture you can

- Sit a 15-mark multiple-choice paper for your option.
- Pick ten short questions and three long questions under the real rules.
- Mark the paper from the lectures, not from memory of the question.

Rules for this drill

Give yourself 20 minutes for Section A and two and a half hours for the rest, or a shorter class test of Section A plus five short questions. Option I candidates ignore the Visual Basic block. Option II candidates ignore the C block. Everyone does the database questions. These questions are original practice in the shape of the 2026 paper. They are not a past paper to memorise.

Option I — Section A

- 1 All C programs contain: A) clrscr B) main C) getch D) start.
- 2 An illegal name is: A) fee B) fee_2 C) 2fee D) Fee.
- 3 % may be applied to: A) int B) a sentence C) a form D) a report.
- 4 k++ is: A) pre-increment B) post-increment C) a comment D) a header.
- 5 Comments use: A) /* */ B) < > C) # ; D) { }.
- 6 continue: A) ends the program B) skips to the next pass of the loop C) declares a variable D) opens a file.
- 7 A for loop is: A) selection B) iteration C) a data type D) a key.
- 8 The translator of a C program is a: A) compiler B) plotter C) bridge D) hub.
- 9 A primary key: A) may repeat B) uniquely identifies a row C) is a form D) is always a date.
- 10 A row is a: A) field B) record C) query D) macro.
- 11 Data stored once and used by many programs shows: A) more unwanted duplication B) integration C) that a DBMS is useless D) a syntax error.
- 12 The object that stores the data is the: A) report B) table C) printer D) label.
- 13 C was designed by: A) Dennis Ritchie B) Blaise Pascal, as the author of C C) the inventor of the mouse only D) a standards body for HDMI.
- 14 In `int a[4]` the first index is: A) 1 B) 0 C) 4 D) 5.
- 15 `scanf` needs: A) the address of the variable B) a plotter C) a foreign key D) a form.

Option I — short questions, answer any ten on the real paper

- i) Write the structure of a C program.
- ii) Explain `#define` with two examples.
- iii) Write a program for the multiplication table of an inputted number.
- iv) Differentiate `break` and `continue`.
- v) Name five reserved words.
- vi) What does `&` do in `scanf`?
- vii) Write a factorial program and show $4! = 24$.
- viii) Differentiate `scanf` and `getchar`.
- ix) Differentiate primary key and foreign key.
- x) If $a = 2$, $b = 5$ and $c = 10$, evaluate $200 * ((a * 10) + c)$.
- xi) Name five components of a DBMS.
- xii) Write a function that returns the greater of two integers.
- xiii) Convert $A = (1/2) \times \text{base} \times \text{height}$ into C.
- xiv) Write a nested loop for the triangle 1 / 12 / 123 / 1234 / 12345.
- xv) Define table, query and report.

Option I — long questions, answer any three

1 Explain switch with a diagram and a program. 2 Explain the types of operators in C. 3 Explain the characteristics of a function, with a program. 4 What is a data model? Explain the types. 5 Explain the data types in MS Access.

Option II — Section A

- 1 The default Step of For...Next is: A) 0 B) 1 C) 2 D) 10.
- 2 A row is a: A) field B) record C) query D) module.
- 3 Printing is the job of a: A) report B) primary key only C) modem D) comment.
- 4 A unique key is the: A) primary key B) caption C) label D) step.
- 5 Dim declares a: A) variable B) cable C) topology D) virus.
- 6 A function: A) cannot return a value B) returns a value C) is a printer D) is a topology.
- 7 Passing a copy is: A) ByVal B) a foreign key C) a report D) OLE only.
- 8 The tree-shaped data model is: A) hierarchical B) relational C) a text box D) SMTP.
- 9 SQL belongs to the: A) hierarchical model only B) relational model C) plotter D) fetch cycle.
- 10 One choice from a group uses: A) an option button B) a label C) a timer D) a bus.
- 11 ADO stands for: A) ActiveX Data Objects B) Analog Data Output C) All Disk Operations D) Automatic Default Option.
- 12 A column is: A) an attribute B) a tuple C) a form D) a macro.
- 13 Case Else in Select Case resembles: A) default B) a primary key C) a hub D) RAM.
- 14 Val is used to: A) convert text to a number B) draw a circle C) shut down Windows D) define a protocol.
- 15 The object that stores records is the: A) table B) report C) label D) cable.

Option II — short and long

Short, any ten: four advantages of a DBMS; three properties of a combo box; declare a Variant; one-to-many with a diagram in words; four Access objects; the structure of Select Case; three duties of a DBA; what an ActiveX control is; full forms of ADO, SQL and DBA; four properties of a text box; a program of even numbers from 0 to 50; sum and average of ten numbers; primary key versus foreign key; three types of error.

Long, any three: arrays and their types; two kinds of loop with examples; a built-in function with a program; data models; data types in MS Access.

Answer key — Section A only

Option I: 1 B, 2 C, 3 A, 4 B, 5 A, 6 B, 7 B, 8 A, 9 B, 10 B, 11 B, 12 B, 13 A, 14 B, 15 A.

Option II: 1 B, 2 B, 3 A, 4 A, 5 A, 6 B, 7 A, 8 A, 9 B, 10 A, 11 A, 12 A, 13 A, 14 A, 15 A.

Mark a short or long answer from the model answer in the lecture named by the topic. A program with the right loop and a missing Dim, or a missing ampersand in scanf, is not a full-mark program. A database answer with no example is not a full-mark answer.